

Analytics Objective

D2S Bharat: From Data-Science Job Demand to Skills, Success and Training Decisions

Team Code5urge · SAS CU Hackathon / Build for Bharat · Draft for team agreement

Problem context. Institutions that prepare data professionals must decide what to teach with limited budget, trainers and seats. They currently lack evidence on three questions: which skills the market *demands and pays for*, which skills are associated with *early-career growth*, and which personal traits are associated with *success among senior* data scientists. The SAS-provided data covers all three: job postings, company-level openings and salaries, and two outcome-labelled employee datasets.

Analytics objective. To identify the technical skills and personality traits most strongly associated with labour-market demand, pay and on-the-job success for data-science professionals, and to translate this evidence into a budget-constrained, explainable training plan that tells an institution *what to teach first*.

Research questions.

RQ	Question	Data	Methods
RQ1 Demand & pay	Which skills and roles are most demanded, and which are associated with higher salary bands and experience requirements?	Analytics Jobs (14,840 postings after cleaning); DataScience Jobs (1,602 rows, 93,005 openings)	Cleaning, skill normalisation to ESCO/O*NET, descriptive statistics, non-parametric tests
RQ2 Junior growth	Which of the five measured skill areas are associated with a high vs. low salary hike among junior data scientists?	JDS Skill Traits (139)	Group tests, correlation, regularised logistic regression, shallow decision tree, repeated cross-validation
RQ3 Senior success	Which Big Five traits are associated with high vs. low success among senior, customer-facing data scientists?	SDS Personality Traits (161)	As RQ2
RQ4 Decision	Given RQ1–RQ3, which training interventions should an institution prioritise under budget and capacity limits?	Evidence from RQ1–RQ3	Constrained optimisation (OR-Tools CP-SAT), sensitivity analysis

Scope and assumptions. Only the SAS-provided datasets are analysed; ESCO and O*NET serve solely as a skill vocabulary. Findings are associations, not causal effects. JDS and SDS are small samples, so model complexity is limited deliberately: with five predictors each, the smaller outcome class gives 13.2 (JDS) and 15.2 (SDS) events per predictor, a favourable ratio, while model complexity stays deliberately constrained. We use L2-regularised logistic regression and decision trees of depth ≤ 3 , repeated stratified k-fold cross-validation, and bootstrap confidence intervals for effects. Salaries are analysed as band midpoints.

Stakeholders and expected outputs.

- **Institutions and curriculum designers:** a ranked, evidence-backed list of skills to teach, and an optimised training plan with its cost–impact trade-offs.
- **Students:** which skills and traits are linked to pay, growth and success.
- **Employers and policy/CSR programmes:** where supply and demand diverge, by role and city.

Success criteria. Each RQ is answered with a quantified finding; predictive models (RQ2, RQ3) are judged against a majority-class baseline using cross-validated accuracy and AUC; and the RQ4 plan states which constraints bind and what extra budget would change.